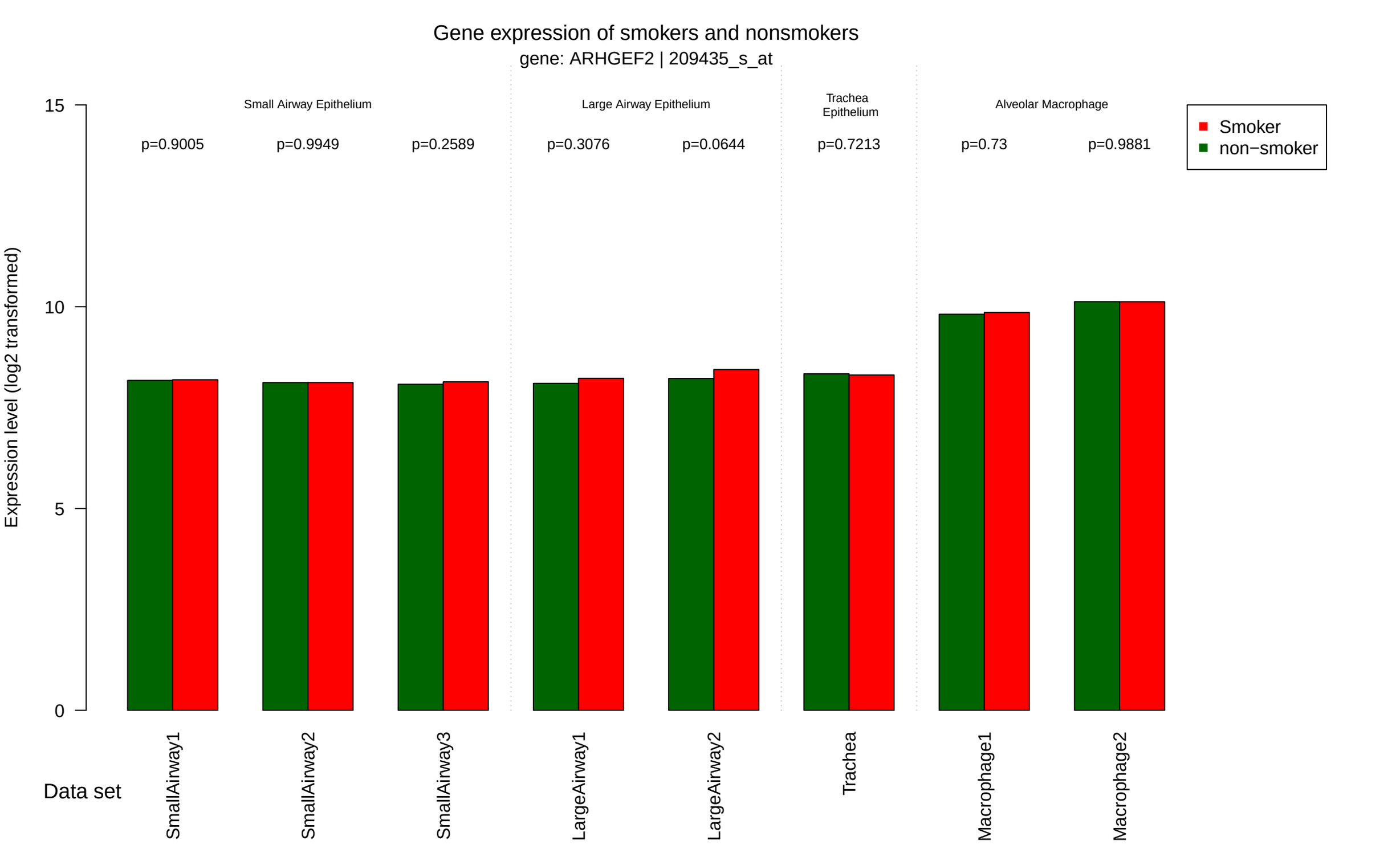
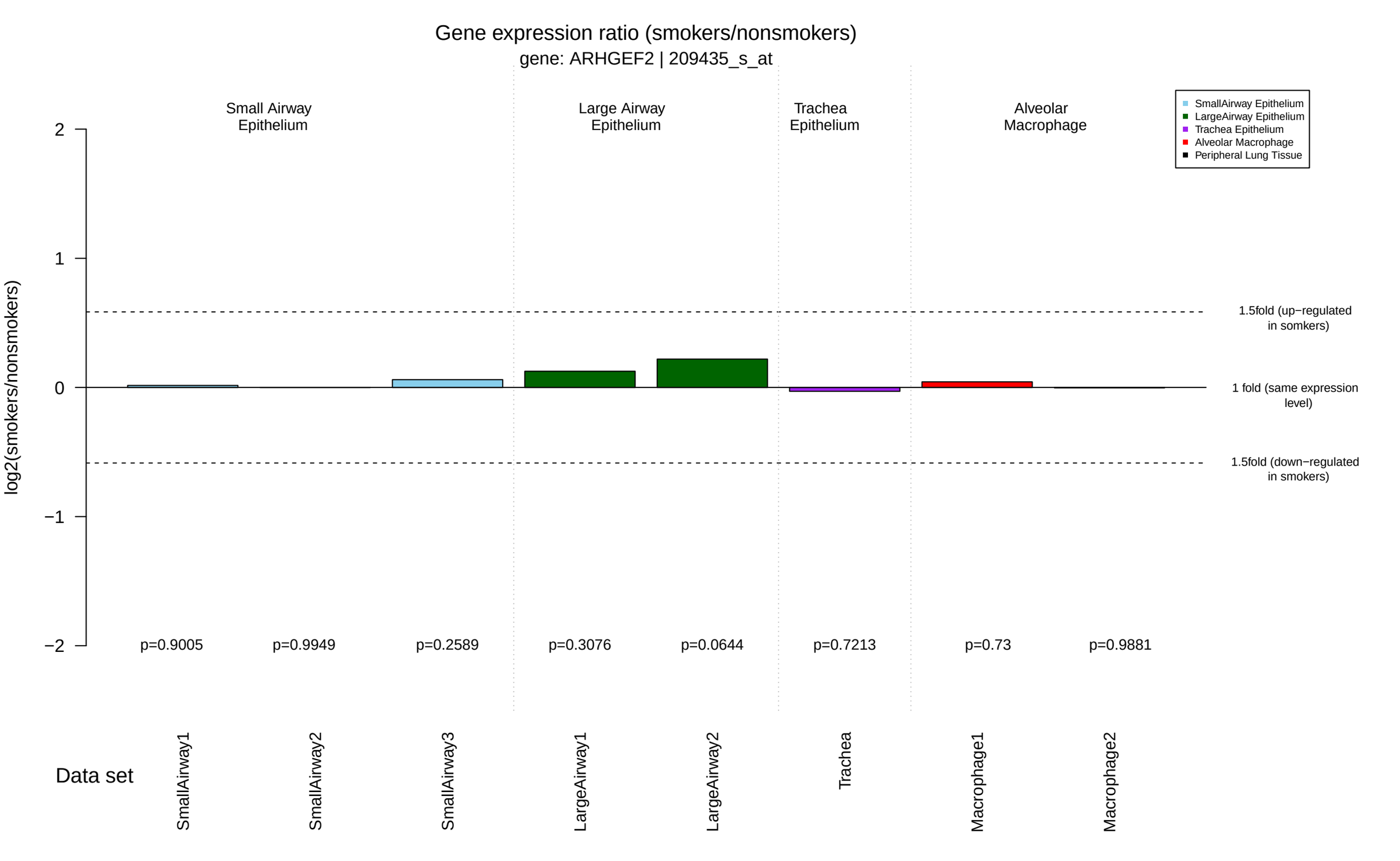
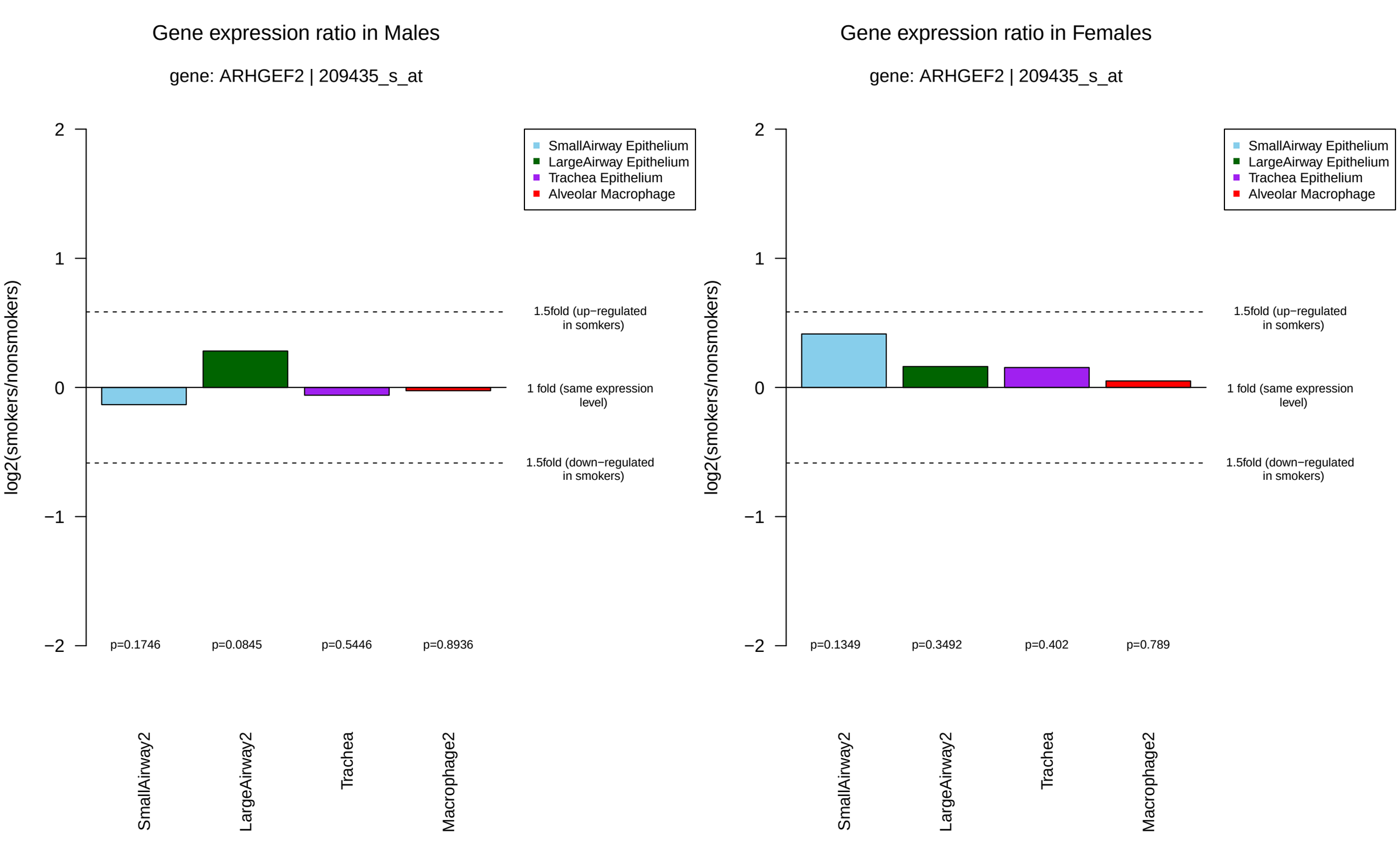
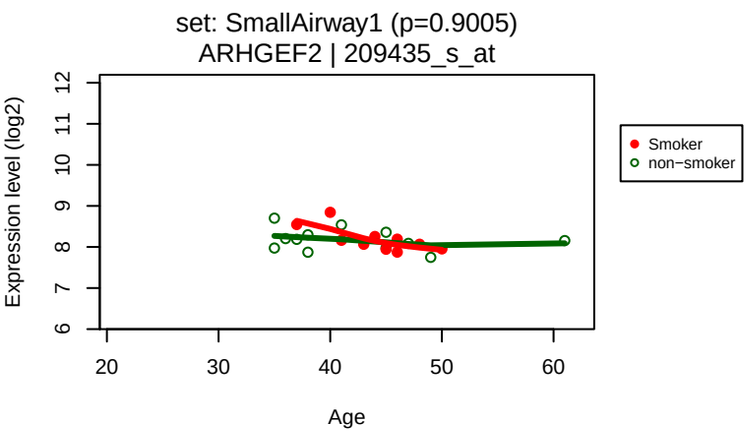
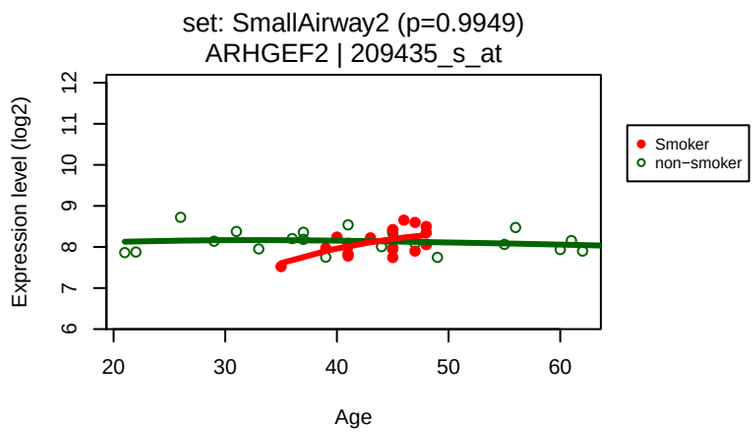
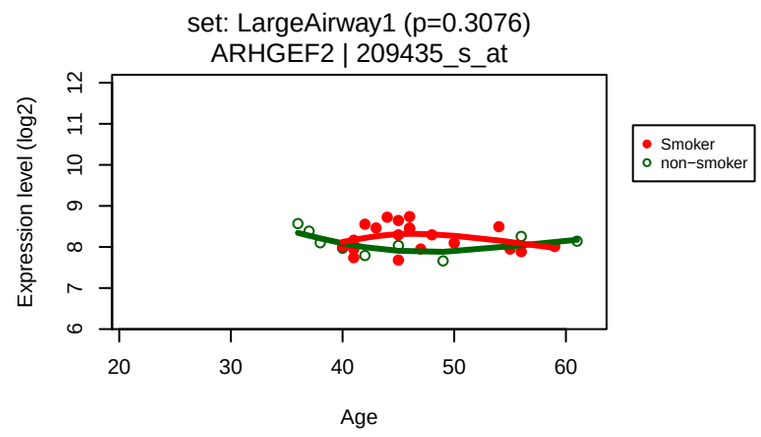

Gene expression of smokers and nonsmokers
gene: ARHGEF2 | 209435_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9005
p=0.9949
p=0.2589
p=0.3076
p=0.0644
p=0.7213
p=0.73
p=0.9881
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

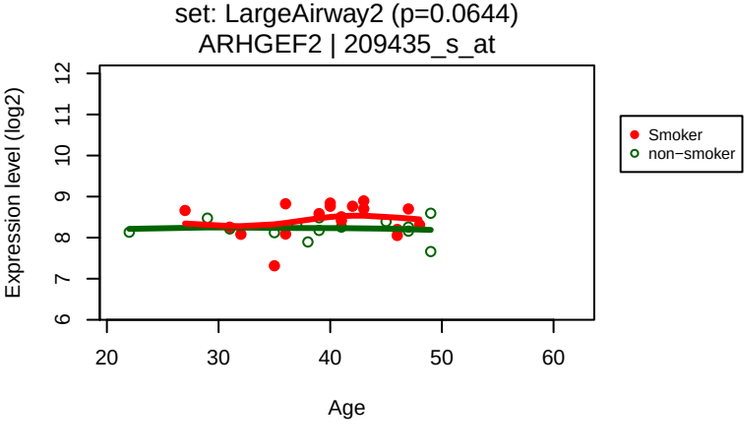

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF2 | 209435_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9005
p=0.9949
p=0.2589
p=0.3076
p=0.0644
p=0.7213
p=0.73
p=0.9881
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

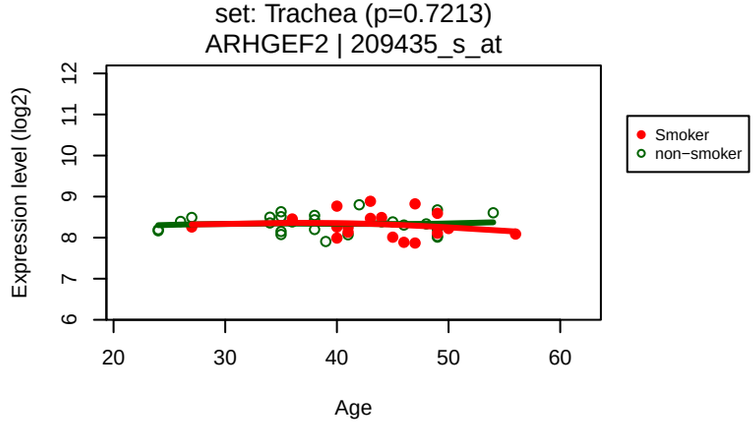

Gene expression ratio in Males
gene: ARHGEF2 | 209435_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1746
p=0.0845
p=0.5446
p=0.8936
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF2 | 209435_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1349
p=0.3492
p=0.402
p=0.789
SmallAirway2
LargeAirway2
Trachea
Macrophage2

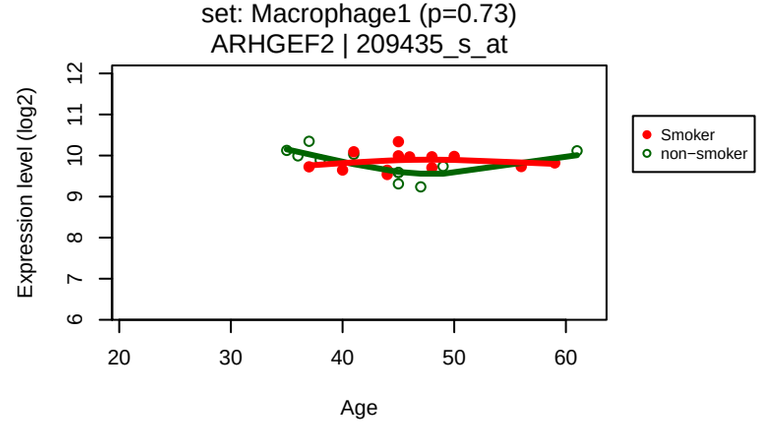

set: SmallAirway1 (p=0.9005)
ARHGEF2 | 209435_s_at
Expression level (log2)
Age
Smoker
non−smoker

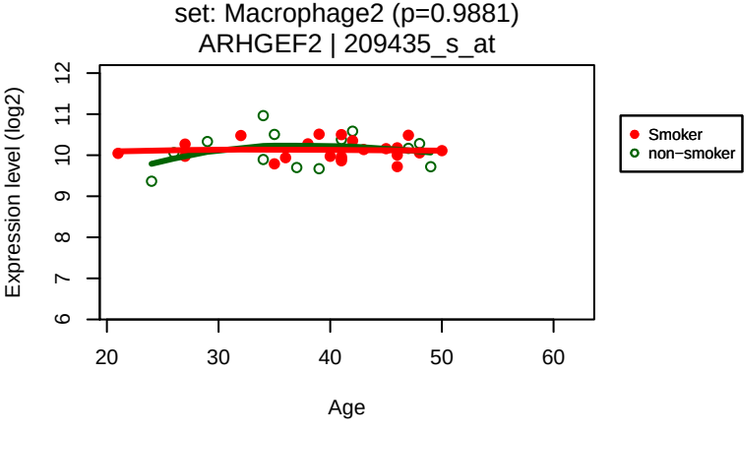

set: SmallAirway2 (p=0.9949)
ARHGEF2 | 209435_s_at
Expression level (log2)
Age
Smoker
non−smoker

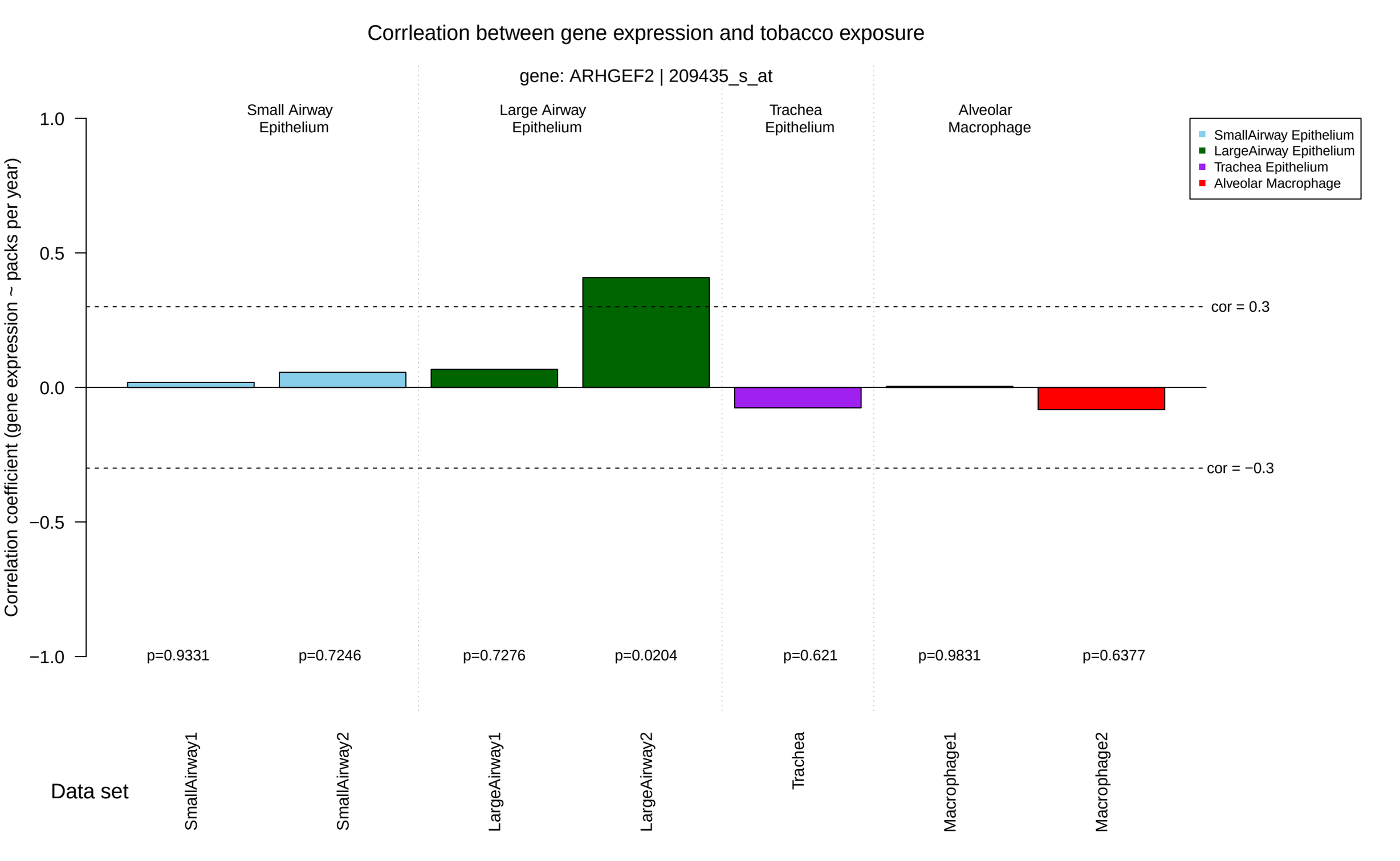

set: LargeAirway1 (p=0.3076)
ARHGEF2 | 209435_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0644)
ARHGEF2 | 209435_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7213)
ARHGEF2 | 209435_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.73)
ARHGEF2 | 209435_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9881)
ARHGEF2 | 209435_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF2 | 209435_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9331
p=0.7246
p=0.7276
p=0.0204
p=0.621
p=0.9831
p=0.6377
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2